

Course: (2026S) COMP-5450-SA - Mobile Programming

Exercise 2: MEMORY MATCHING FLUTTER GAME

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GitHub URL: <https://github.com/achatessaran/Memory-Matching-App>

Project Overview

This project is a Memory Matching Game developed using Flutter and Dart. The application challenges users to match pairs of shape cards using memory skills.

The game includes:

- Welcome screen
- Interactive memory card gameplay
- Timer functionality
- Move counter
- Pair counter
- Best score tracking
- Restart functionality
- Home navigation
- Custom app launcher icon
- Responsive 20-card game layout

The application was tested successfully on Android Emulator.

Technologies Used

- Flutter
- Dart
- Android Studio
- Android Emulator (Pixel 6)
- GitHub

Features

1. Welcome Screen

- Displays game title
- Start Game button
- User-friendly interface

- Gradient background design

2. Memory Matching Gameplay

- Shape image cards
- Randomized card arrangement
- Flip and match logic
- 20-card responsive layout (4 columns × 5 rows)

3. Score Tracking

- Move counter
- Pair counter
- Game timer
- Best score display

4. Navigation

- Home button
- Restart game button

5. Winning Dialog

- Displays moves taken
- Displays time taken
- Play Again option

6. App Design Improvements

- Rounded card corners
- Improved shadows and UI styling
- Responsive layout design
- Custom launcher icon

Project Structure

memory_matching_game

```
|  
|— android  
|— ios  
|— lib
```

```
| └── main.dart
|
| └── assets
|     ├── images
|     ├── red_circle.png
|     ├── blue_square.png
|     ├── green_triangle.png
|     ├── orange_diamond.png
|     ├── purple_star.png
|     ├── pink_heart.png
|     ├── yellow_hexagon.png
|     ├── cyan_oval.png
|     ├── brown_pentagon.png
|     └── teal_cross.png
|
| └── shapes.png
| └── pubspec.yaml
| └── README.pdf
```

How to Run the Project

Step 1: Install Flutter SDK and Android Studio.

Step 2: Open the project in Android Studio.

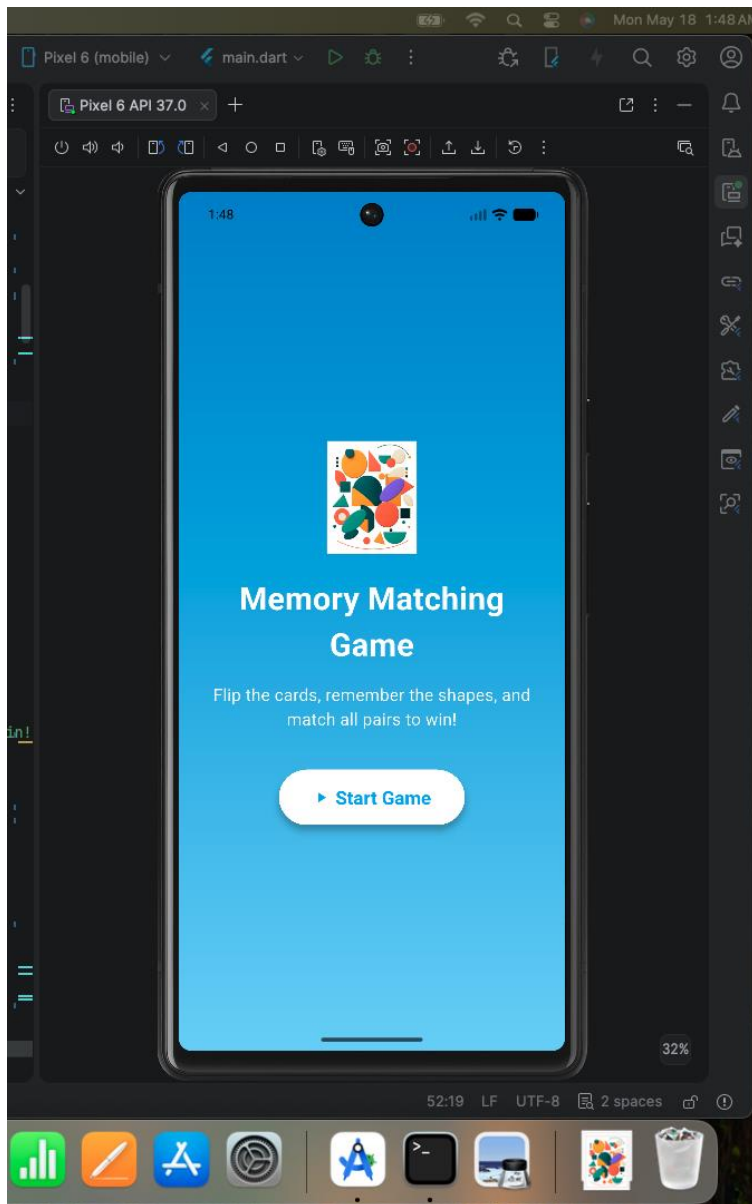
Step 3: Run the command: `flutter pub get`

Step 4: Start Android Emulator.

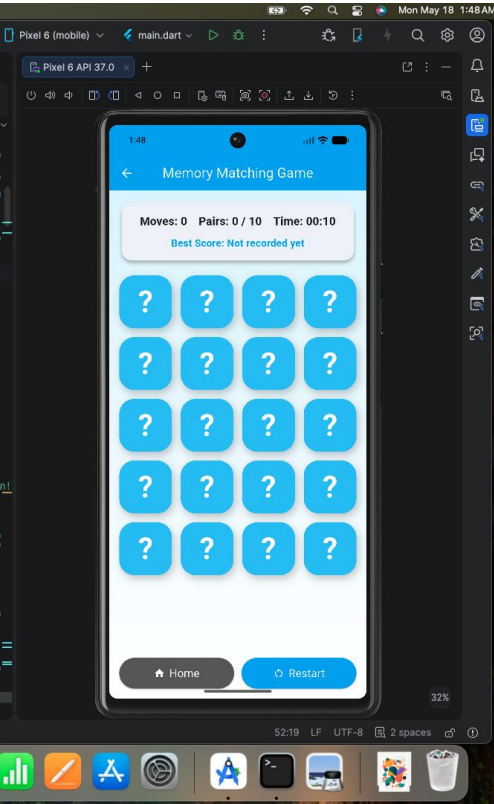
Step 5: Click Run ► in Android Studio

Screenshots

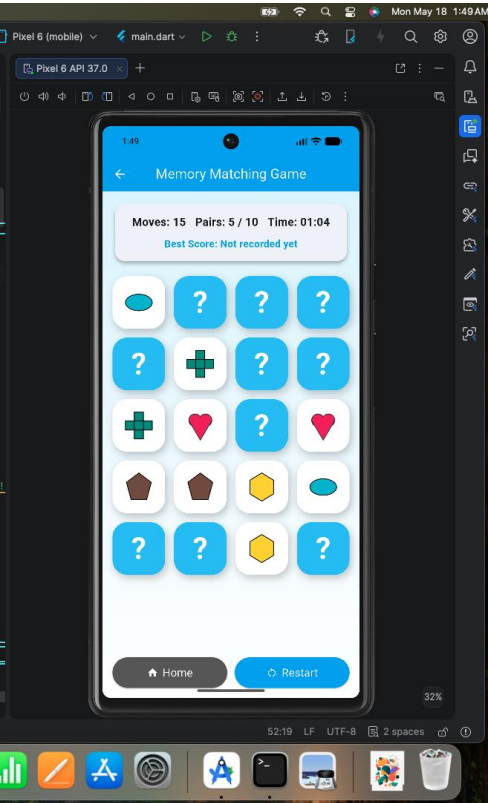
Welcome Screen



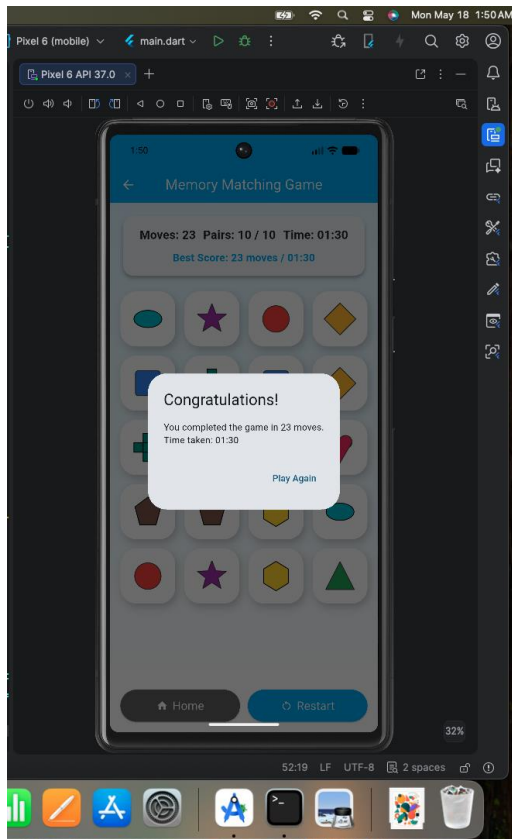
Game Opening Screen



Matching Gameplay



Winning Dialog



App Icon



Conclusion

This project demonstrates the implementation of a responsive Memory Matching Game using Flutter and Dart. The application includes interactive gameplay, responsive design, score

tracking, timer functionality, navigation features, custom launcher icon integration, and a professional user interface.